RESEARCH PAPER APPENDIX / OPERATIONS MANUAL

NFC Keycard Access Control System

Operational User Guide for Smartphone & Dedicated IoT Hardware Terminals

SYSTEM TARGET	BACKEND ARCHITECTURE	FREQUENCY BAND	DOCUMENT STATUS
Home-Spot MJP Residences	Supabase Cloud Serverless API	13.56 MHz (ISO 14443 Type A)	Published / Capstone Verified

1. System Architecture & Core Flow

The Home-Spot Contactless Gate Access Control System provides physical premise security and attendance tracking across MJP Residences. The platform features an automated dual-modality architecture that processes contactless scans from both mobile devices and microcontrollers using a centralized cloud endpoint.

> **Centralized Gate REST API**
> **Endpoint:** `POST https://jfeosjevabaadudestcf.supabase.co/functions/v1/gate-tap`
> **Headers:** `Content-Type: application/json` • `apikey: <SUPABASE_PUBLIC_KEY>`
> **Request Payload:** `{ "cardUid": "NFC-101", "source": "NFC_KIOSK" }`

State Transition & Automatic Direction Resolution

The system resolves scan direction dynamically through server-side state evaluation:

- **Initial Registration Tap:** When a tenant taps for the very first time without prior tap records, the direction is guaranteed to resolve to `IN` (Entry).
- **Subsequent Scans:** If currently recorded `INSIDE`, next tap resolves to `OUT` (Exit). If recorded `OUTSIDE`, next tap resolves to `IN`.
- **Curfew Detection:** Any `IN` tap recorded between **10:00 PM and 5:00 AM (Asia/Manila)** is permanently tagged with `Remarks: Curfew Violation` and triggers an automated notification.

2. Modality A: Smartphone-Based NFC Terminal (Soft Kiosk)

The Smartphone Terminal leverages modern commercial mobile devices equipped with Near Field Communication (NFC) chips. It serves as a lightweight, portable scanning station suitable for security guards on patrol or an unassisted tablet kiosk at the front lobby.

2.1 Hardware & OS Prerequisites

- **Supported OS:** Android 8.0+ (with Google Play Services) or Apple iOS 14.0+ (iPhone 7 and newer).
- **Connectivity:** Active Internet connection via building Wi-Fi or 4G/5G mobile broadband.
- **Software Environment:** Modern Chromium browser (supporting Web NFC API) or third-party NFC automation utility (e.g., NFC Tools Pro, Apple Shortcuts).

2.2 Step-by-Step Operator Instructions

1. **Enable NFC Subsystem:** On the smartphone, navigate to *Settings → Connected Devices → Connection Preferences* and toggle **NFC** to *ON*.
2. **Mount or Position Terminal:** Place the device securely in the lobby kiosk stand with the back antenna clear of metallic obstructions.

3. **Tenant Keycard Presentation:**
    - Instruct the tenant to hold their issued keycard flat against the phone's NFC detection zone (upper-rear quadrant for iOS, center-rear for Android).
    - Maintain contact for a minimum dwell time of **400 milliseconds**.
4. **Observe Real-Time Acoustic & Visual Feedback:**
    - **Entry (IN):** Terminal produces a dual high-frequency harmonic chime (880 Hz → 1174 Hz) and displays a green entry confirmation badge.
    - **Exit (OUT):** Terminal sounds a distinct descending chord (659 Hz → 523 Hz) and displays an exit acknowledgment.

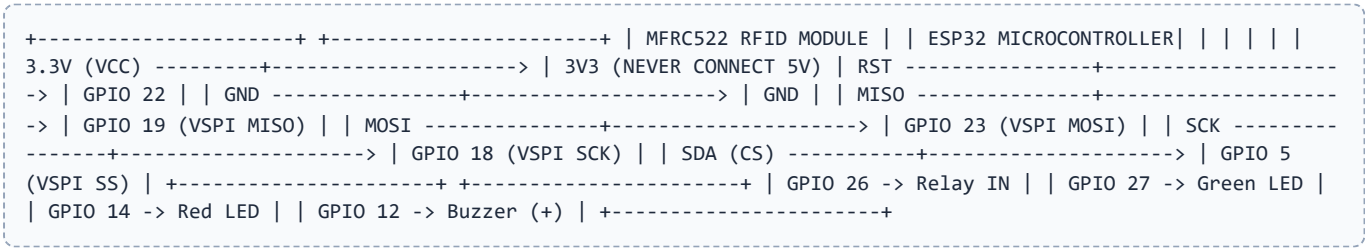
3. Modality B: Dedicated IoT Hardware Terminal (Automated Gate Turnstile)

The dedicated hardware terminal is an embedded IoT appliance designed for permanent installation on turnstiles, perimeter gates, and magnetic strike door locks. It functions autonomously without human supervision.

3.1 Bill of Materials (BOM) & Technical Specifications

Component	Model / Parameter	Functional Role
Microcontroller	ESP32-WROOM-32 (38-pin DevKit)	Embedded execution, Wi-Fi 802.11 b/g/n, TLS 1.3 HTTP client
Contactless Reader	MFRC522 / PN532 (13.56 MHz)	Contactless ISO/IEC 14443 Type A UID interrogation
Actuator Interface	5V Single-Channel Relay (Optocoupled)	Switches 12V DC power to solenoid strike lock (500mA rating)
Visual Indicators	Bi-Color 5mm Common-Cathode LED	Green: Authorized Entry; Red: Access Denied / Unassigned Card
Acoustic Transducer	5V Active Piezo Buzzer (2.3 kHz)	Auditory transaction feedback for blind accessibility
Power Subsystem	12V 2A DC PSU + LM2596 Step-Down	Provides 12V to lock solenoid and stepped-down 5V to ESP32

3.2 Wiring Topology & Microcontroller Pinout (SPI Bus)



3.3 Firmware Embedded Operation Cycle

1. **Standby State:** ESP32 continuously polls the RF field at 10 Hz. Status LED pulses amber/standby.
2. **Card Induction:** When an NFC transponder enters the 0–40mm induction field, the microcontroller extracts the raw Hexadecimal Unique Identifier (UID).
3. **Secure API Dispatch:** The firmware dispatches an HTTPS POST payload to the cloud edge function with a 2000ms watchdog timeout.
4. **Access Decision & Actuation:**
   - **HTTP 200 (Granted):** Energizes GPIO 26 (Relay) for exactly **5.0 seconds** allowing physical passage; pulses Green LED and double-beeps the buzzer.
   - **HTTP 400/404 (Denied/Pending):** Latches relay shut; flashes Red LED 3 times and sounds a continuous 1.5-second error buzz.

4. Diagnostic Matrix & Comparative Analysis

Observed Symptom	System Root Cause	Remediation Procedure
HTTP 400: Keycard Pending Issuance	Tenant record exists but room assignment is still pending admin approval.	Admin must open *Admin Dashboard → Tenants* and assign an unoccupied unit before the card can unlock the gate.
HTTP 404: Unrecognized Card	Card UID not enrolled in `tenants` collection.	Verify tenant's card UID in the database or register the physical transponder via Admin Dashboard.
Hardware Timeout / Fail-Safe	Wi-Fi disconnect or gateway failure.	ESP32 firmware fails closed (locked); auto-reconnects to the network every 10 seconds.

Comparative Evaluation for Capstone Defense

Evaluation Metric	Smartphone Terminal (Modality A)	Dedicated Hardware Terminal (Modality B)
Target Deployment	Guard patrol, front reception, mobile checkpoint	Unattended perimeter turnstile, automated security gate
Physical Actuation	Manual release by guard upon audible chime	Autonomous 12V solenoid lock actuation via relay
Cost Overhead	₱0.00 (utilizes existing staff smartphone)	Low component cost (~₱850–₱1,100 total BOM)
Mean Response Latency	~320 milliseconds (TLS over 4G/Wi-Fi)	~380 milliseconds (TLS over 2.4 GHz Wi-Fi)